

# Never Overcharge Batteries

Overcharging batteries can damage them irretrievably

Signs of overcharging:- excessive water loss in flooded batteries requiring frequent topping up, continuously warm batteries, bulging container

[www.relicellbattery.com](http://www.relicellbattery.com)



VRLA and Tubular batteries

VRLA Batteries - 12V 7AH to 250 AH  
Tubular Batteries - 12V 40AH to 210 AH

Part of a Rs.1000 crore IT industry group  
Tested and certified by Central Power Research Institute, Bangalore (CPRI) and Electronic Test and Development Centre (STQC Directorate, Ministry of Information Technology, Govt. of India). Download Test Certificates at [www.relicellbattery.com](http://www.relicellbattery.com)  
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VRLA Batteries approved and registered in DGS&D for government purchases  
47 Service Centres across India

## RELICELL

Greenvision Technologies Pvt.Ltd

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Email: [info@relicellbattery.com](mailto:info@relicellbattery.com)

## Top 40 Engineering Colleges

Thank you for including our college in Top 40 Engineering Colleges ranking that appeared in May issue. We suggest you to include the total marks scored by each institute in the list. That would help us assess our position quantitatively vis-à-vis other institutes. Quantified weightages for various parameters would also be helpful to the participating institutions in evaluating themselves.

Principal  
Shri Ramdeo Baba  
Kamla Nehru Engineering College  
Nagpur

The rank secured by our college amongst Top 40 Engineering Colleges has boosted our faith in the pursuit of excellence. We express our good wishes to your entire team involved in the survey and hope you will continue with this activity with more zeal for the benefit of the student community.

Secretary

SDME Society, Dharwad

EFY: Thank you notes and suggestions have also been received from many other institutes including B.N.M. Institute of Technology, Bengaluru and Textile & Engineering Institute, Ichalkaranji, Kolhapur.

## PCBs using Express PCB

Regarding the 'Developing PCBs using Express PCB' article published in February issue, I would like to mention that you can use photo paper as well as normal plain paper. When transferring the conductor track, after ironing the paper, you have to put it into water. It requires two to five minutes to remove the photo paper from the board. If you use normal paper, after 10 minutes in water you need to remove the paper gently and scrub slowly by hand. The author has mentioned that the ink won't transfer because of sharpness, which is not correct. It won't transfer by heat even though there is paper available in the market to transfer the ink printed.

Zubair S.S.  
Through e-mail

The author **Somnath Chakraborty** replies:

I agree that a photo paper may be used, but it may not be available always. So you can easily use a plain paper. After ironing, directly pull away the paper. There is no need to wash it under water separately because when you etch it with  $\text{FeCl}_3$  solution, the excess paper attached to the board is automatically removed.

And the inkjet printer should not be used to print out the circuit because the ink from the paper will not transfer to the PCB by heat. Only carbon from the laser printer transfers to the PCB while heating. But if you use heat-transferable inkjet printer paper, it will be alright.

## Electronic Lock

In the 'PIC Microcontroller-based Electronic Lock' article of June issue, on page 72, in third column, please read the second paragraph as "The advantage of this system is that the password set by the user is stored in the EEPROM of PIC16F877A and hence it is not lost when the supply is switched off. When the supply is switched on again, the system retains the new password."

Anshuman Bezborah  
The author

## Analogue Clock

In the 'Analogue Clock' article published in May issue, please note that:

1. In the schematic diagram (Fig. 4), NOR gate N1 pins 5 and 6 of IC5 (CD4001) should be connected to pin 11 of IC4 (CD4033) instead of pin 12 as shown.

2. In the same figure, NOR gate N2 pin 1 of IC5 (CD4001) should be connected to pin 12 of IC4 (CD4033) instead of pin 11 as shown.

3. These corrections are to be done in the PCB (Fig. 6) accordingly by cutting the copper tracks and connecting with jumpers wires.

4. In the PCB layout again, the 6V battery polarity symbols should be interchanged. That is, positive terminal of the battery should be connected to resistor R85.

B. Shanker Rao  
The author